

Course Syllabus: Systems Analysis and Design 1st Semester, A.Y. 2026–2027

Course Information

Instructor: [SAD Instructor Name] | Units: 3 | Schedule: TTh 3:00–4:30 PM | Room: IT Lab 3

1. Course Description

Systems Analysis and Design (SAD) provides students with a structured approach to designing and developing information systems. The course covers the Systems Development Life Cycle (SDLC), requirements gathering, process modeling using Data Flow Diagrams (DFDs) and Use Case Diagrams, database design, prototyping, and system implementation. Students will work in teams to analyze a real-world problem and produce a complete system design proposal.

2. Learning Outcomes

- LO 1: Apply the SDLC methodology to plan and manage a software development project.
- LO 2: Elicit and document system requirements through interviews, surveys, and observation.
- LO 3: Create process models using DFDs, flowcharts, and UML diagrams.
- LO 4: Design a functional prototype of an information system based on user requirements.

3. Weekly Schedule

Week	Topic	Assessment
1	Introduction to Systems Analysis; SDLC Overview	Quiz 1
2	Agile vs Waterfall; Project Management Basics	Discussion
3	Requirements Elicitation; Interviews and Surveys	Lab 1: Requirements Doc
4	Use Case Diagrams; Actor Identification	PT 1: Use Case Diagram
5	Data Flow Diagrams; Context Diagrams; Level 0 and 1 DFDs	Lab 2
6	Process Modeling; Decision Tables and Trees	Quiz 2
7	Data Modeling; ER Diagrams; Data Dictionary	Lab 3
8	Midterm Examination	Midterm Exam
9	System Design; Input/Output Design; UI Wireframes	PT 2: Wireframes
10	Database Design; Normalization Review	Lab 4
11	Prototyping; Low-Fidelity and High-Fidelity Prototypes	Lab 5
12	System Testing; Test Plans and Test Cases	PT 3: Test Plan

13	Implementation Planning; Conversion Strategies	Lecture
14	System Maintenance; Change Management	Quiz 3
15	Group Presentations: System Design Proposal	Group Presentation
16	Final Examination	Final Exam

4. Grading System

Component	Weight
Quizzes / Participation	15%
Laboratory Outputs	25%
Performance Tasks	20%
Midterm Exam	20%
Final Exam	20%

5. Policies

Team Work: All group members must contribute equally. Division of labor must be documented and submitted alongside group outputs.

Submission: All outputs must be submitted via the LMS before 11:59 PM on the due date. Email submissions are not accepted.

Consultation: Students are encouraged to schedule consultations during the instructor's office hours for project feedback.

6. References

1. Kendall, K. & Kendall, J. (2019). Systems Analysis and Design. Pearson.
2. Sommerville, I. (2016). Software Engineering. Pearson.
3. Dennis, A. et al. (2021). Systems Analysis and Design: An Object-Oriented Approach with UML. Wiley.